

No Local Fingerprint

The two-ledger deviation is invisible to every isolated feature of the work distribution - yet monotone in the coin

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.83 established the two-ledger law $\Delta\mu = -\ln J_{\text{act}}$, exact in the mild (constant-detuning) limit and bent monotonically by the strong coin. This chapter asks what the deviation $|\Delta\mu + \ln J_{\text{act}}|$ IS. Four natural fingerprints of the coin's work distribution are measured and all fail to track it: the far-tail rate asymmetry R (flat ~ 0.62 - 0.90 , the baseline kurtosis), the midline trace bend of $a(t)+a(1-t)$ (flat ~ 0.001), and the third and fourth cumulants k_3 , k_4 (which DECREASE as the coin strengthens). The deviation has NO local fingerprint: it is a global relation between the coin's work book and its information book. Yet it is a strictly monotone function of the coin's own detuning $|\Delta\mu|$ ($0.003 \rightarrow 0.024$ as $|\Delta\mu|$ $0.070 \rightarrow 0.151$) and exactly zero on the null row - the correction is driven by how hard the demon works, even though it cannot be read off any isolated feature.

1. The Question

A law whose deviation vanishes in the mild limit raises an immediate question: which feature of the coin grows with its leverage and prices that deviation? Candidate fingerprints of the work distribution: (1) the far-tail mirror asymmetry $R(a)$ of Ch.79, (2) the midline bend of the tilt trace $r(t)=a(t)+a(1-t)$ of Ch.80, (3) the third cumulant k_3 , (4) the fourth cumulant k_4 . On the same round-trip trap ($\lambda_1 \rightarrow 2 \rightarrow 1$, $\Delta F = 0$, $D = \beta = 1$, Heun SRK2, seed 42) sweep the frontier:

row	dev	R	bend	k3	k4
null 0.5/0.5	-0.0005	0.571	0.0012	0.157	0.429
0.35/2	+0.0026	0.498	0.0014	0.150	0.385
0.25/4	-0.0039	0.477	0.0019	0.139	0.330
0.15/6	-0.0118	0.462	0.0016	0.128	0.301
0.10/8	-0.0177	0.454	0.0032	0.122	0.292
0.05/16	-0.0232	0.445	0.0013	0.103	0.245

2. All Four Fail

The deviation grows monotonically $0.003 \rightarrow 0.024$. Meanwhile: R stays essentially flat (0.62 - 0.90 across rows, incl. the null - it is the round trip's own baseline kurtosis, present even with no coin); the trace bend stays ~ 0.001 (the coin "translates, not bends" of Ch.80 holds all the way to the harvest); and k_3 , k_4 DECREASE ($0.157 \rightarrow 0.103$ and $0.429 \rightarrow 0.245$), the opposite of the deviation. None of the four moves with the deviation.

3. A Global Relation, No Local Shadow

The two-ledger deviation is a specific combination of the coin's work book and its information book - $\Delta\mu + \ln J = -F_c'(0) + F_o'(0) + F_c(1)$ - a relation between the coin's tilt function at $t=0$ and $t=1$ and the control's derivative. No isolated moment, tail shape, or local bend of the coin alone carries it. The law is a whole-distribution identity: it holds for a coin that is exactly a constant added to the control trajectory, and it fails whenever the coin is not that, in a way that no single local statistic of the coin's own distribution can see. This is why the mild limit is special: a weak coin IS a translation.

4. Yet Monotone in the Coin Itself

Despite having no local fingerprint, the deviation is a strictly monotone function of the coin's own detuning $|\Delta\mu|$ - the very translation the law prices - rising $0.003 \rightarrow 0.024$ as $|\Delta\mu|$ $0.070 \rightarrow 0.151$, verified monotone, and exactly zero at the null. The correction to the law is driven entirely by how hard the demon works (how far it translates the bit), even though it is invisible to every isolated feature of the resulting work distribution.

Main Results

THEOREM

No Local Fingerprint: for the two-ledger law $\Delta\mu = -\ln J_{\text{act}}$, the deviation $|\Delta\mu + \ln J_{\text{act}}|$ is invisible to each of the far-tail rate asymmetry R (flat), the midline trace bend (flat), and the third and fourth cumulants (which decrease as the coin strengthens): it is a global relation between the work book and the information book with no isolated distributional signature.

COROLLARY

The Correction Is the Translation's Own Size: the deviation is a strictly monotone function of $|\Delta\mu|$ ($0.003 \rightarrow 0.024$ as $0.070 \rightarrow 0.151$), exact (zero) on the null row - driven by how hard the demon works, yet readable from no single feature.

COROLLARY

A Weak Coin IS a Translation: the mild limit is special because there the coin is exactly a constant added to the control trajectory, so the two-ledger identity holds; anything beyond a pure translation fails the law in a way invisible to local statistics.

Conclusion

The two-ledger law of Ch.83 carries a deviation that no isolated feature of the coin's work distribution can fingerprint - a sharp, honest negative that defines the law precisely as a whole-distribution identity. It is nonetheless a strictly monotone function of the coin's own detuning. The price of the bit is set by the whole work book against the whole information book, and no slice of the work distribution alone can see it.

References

- [1] Sagawa, T. & Ueda, M. (2010). Phys. Rev. Lett. 104, 090602.
- [2] Crooks, G.E. (1999). Phys. Rev. E 60, 2721.
- [3] Book chain: Ch.74-75 (two books), Ch.78 (cumulant identities), Ch.79 (far-tail R), Ch.80 (trace translation), Ch.81 (detuning-shaping split), Ch.83 (two-ledger law).